

Project Demonstration Phase:

Import, Transform Maps & Reporting in ServiceNow

1. Executive Summary & Project Overview

This project demonstration document outlines the full end-to-end implementation lifecycle for ingesting bulk employee data from an external spreadsheet into the ServiceNow platform. Using **System Import Sets**, **Transform Maps**, **Coalesce Constraints**, **Analytics Reporting**, and **Dashboards**, this solution simulates an enterprise-grade migration scenario where human resources and administrative data must be imported, cleansed, updated, and visualized securely.

Key Objectives:

- Structure raw spreadsheet data into a standardized .xlsx format.
- Establish custom target tables (Employee Test) within ServiceNow.
- Stage, transform, and load records using Import Sets and Field Mappings.
- Prevent data redundancy and handle updates intelligently via Coalesce fields.
- Build actionable visualizations (Pie charts, Bar charts, and List reports).
- Consolidate metrics into a role-restricted interactive dashboard.

2. Phase-by-Phase Demonstration & Execution Architecture

Phase 1: Data Preparation (.xlsx Source Generation)

Raw data is structured in a standard spreadsheet application to prepare it for ingestion.

- **Schema Definition:** Employee ID (SB-0001 to SB-0015), Name, Email, Department, Location.
- **Export Format:** Exported from Google Sheets as Sample Spreadsheet.xlsx.

Phase 2: Custom Target Table Creation

Establishes the permanent destination repository within ServiceNow.

- **Navigation:** Tables > Create New
- **Table Label:** Employee Test
- **Table Name:** u_employee_test
- **Form Layout Config:** Added custom string fields via Form Layout:
 - Employee ID (u_employee_id)

- Employee Name (u_employee_name)
- Email (u_email)
- Department (u_department)
- Location (u_location)

Phase 3: Staging Area via Import Sets

Handles raw file ingestion before data hits target tables.

- **Navigation: Load Data**
- **Import Set Table Label:** Employee Import (u_employee_import)
- **Source Type:** File Upload (Sample Spreadsheet.xlsx)
- **Execution Result:** Progress log confirmed State: Complete and Completion code: Success with 15 initial records inserted.

Phase 4: Transform Map Configuration & Execution

Maps incoming staging attributes to target table columns.

- **Transform Map Name:** Sample Spreadsheet Import
- **Source Table:** u_employee_import
- **Target Table:** u_employee_test
- **Mapping Strategy:** Configured via **Mapping Assist** and **Auto Map Matching Fields** to pair fields (u_name → u_employee_name, u_employee_id → u_employee_id, etc.).
- **Transformation Execution:** Processed 15 inserts with 0 errors.

Phase 5: Verification & List Customization

- **Navigation:** Opened Employee Test table directly from the Application Navigator.
- **List Personalization:** Rearranged columns using **Personalize List Columns** to verify structural integrity and data accuracy across all 15 records.

3. Advanced Governance: Coalesce & Duplicate Prevention

To handle recurring bulk imports without creating redundant rows, Coalesce is leveraged.

Mechanism & Implementation Steps:

1. Navigate to **System Import Sets > Transform Map**, and open Sample Spreadsheet Import.
2. Access the **Field Maps** related list.
3. Toggle the **Coalesce** attribute to true on the unique identifier field (u_employee_id).
4. **Re-import Test Execution:** Uploaded a modified dataset containing 4 records (2 existing modified records with updated names/emails, and 2 brand-new records).
5. **Transform History Analysis:**
 - **Inserts:** 2 (New unique rows)

- **Updates:** 2 (Existing rows modified via Coalesce key match)
- **Ignored/Skipped:** 0 (or 4 when re-uploading identical files without changes, proving idempotency).

4. Analytics Reporting Implementation

Three distinct reports were created under **Usage and Governance > Reports** to analyze employee data distribution:

1. **Employees by Department (Pie Chart):**
 - **Table:** Employee Test (u_employee_test)
 - **Type:** Pie Chart
 - **Group By:** Department
 - **Aggregation:** Count
2. **Employees by Location (Bar Chart):**
 - **Table:** Employee Test (u_employee_test)
 - **Type:** Bar Chart
 - **Group By:** Location
 - **Aggregation:** Count
3. **Employee List Report (List View):**
 - **Table:** Employee Test (u_employee_test)
 - **Type:** List
 - **Columns Displayed:** Employee ID, Employee Name, Email, Department, Location.

5. Dashboard Consolidation & Security Controls

Dashboard Creation:

- **Backend Access Configuration:** Adjusted Access Control List (pa_dashboards) to enable **Admin Overrides** (true), permitting custom dashboard creation.
- **Dashboard Name:** Employee Analytics Dashboard (pa_dashboards.form).

Widget Integration & Sharing:

- Added all three reports (Employees by Department, Employees by Location, and Employee List Report) to the dashboard grid via the Share and Add to Dashboard utility.
- Configured user, group, and role access restrictions to secure executive visibility.

6. Project Conclusion & Takeaways

1. **Automated Efficiency:** Replaced manual data entry with a repeatable, automated pipeline using ServiceNow Import Sets and Transform Maps.
2. **Data Integrity via Coalesce:** Enabled enterprise-grade deduplication, ensuring recurring uploads update existing records instead of corrupting databases.
3. **Actionable Insights:** Consolidated raw data into visual dashboards for leadership teams, empowering data-driven decision-making.